An infrared triangle for quantum spin chains

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The infrared triangle of gauge theory—soft theorem, asymptotic symmetry, memory effect—is transplanted to quantum spin chains. Proved: a truncated symmetry acts only at its endpoints, its charge algebra centrally extended by the topological index; in the bilinear isotropic ferromagnet, on its regular two-magnon domain, a soft magnon exits with universal phase slope; granted a scattering hypothesis, a quantized wall step per transmitted magnon. Its edges stay conjectures. The continuum rule that memory is the zero-frequency soft factor fails: the soft factor is a phase, the memory a charge.

A ferromagnetic spin chain contains two objects that can be pictured without formalism: the *magnon*, a single overturned spin spreading as a wave, and the *domain wall*, the boundary between a region of up spins and a region of down spins. This Letter concerns what each does as the magnon's momentum is taken small, and the bookkeeping that controls both. The whole result in one sentence: on the lattice, a soft magnon leaves two-body ferromagnet scattering with a universal phase slope; the symmetry behind this acts at the ends of the chain as a concrete algebra of operators on matrix-product bond space; granted the scattering hypothesis H-AD-G stated below, a magnon that crosses a domain wall displaces it by a quantized charge step—and the continuum rule that identifies the memory with the zero-frequency limit of the soft factor is *false* here: what survives, and is proved, is a current-flux identity and charge bookkeeping—charge transport, not phase accumulation.

In continuum field theory, soft theorems, asymptotic symmetries, and memory effects form the *infrared triangle*, three faces of one Ward identity, established for photons and gravitons at null infinity [1, 2]. Hamada and Sugishita extended the triangle to Goldstone bosons of a broken global symmetry [3]; scalar and fracton versions followed [4–6], with local derivations avoiding null infinity [7] and a careful account of infrared finiteness [8]. Each corner has a condensed-matter ancestor: soft-magnon decoupling and Adler zeros of nonrelativistic Goldstone theories are by now systematic [9–12]; Lan and Xiao described magnon-driven domain-wall displacement as a two-body collision [13, 14]. Missing is a lattice statement at finite spacing, no continuum limit; we supply one, prove what we can, label the rest conjecture, and report one refutation as a central result.

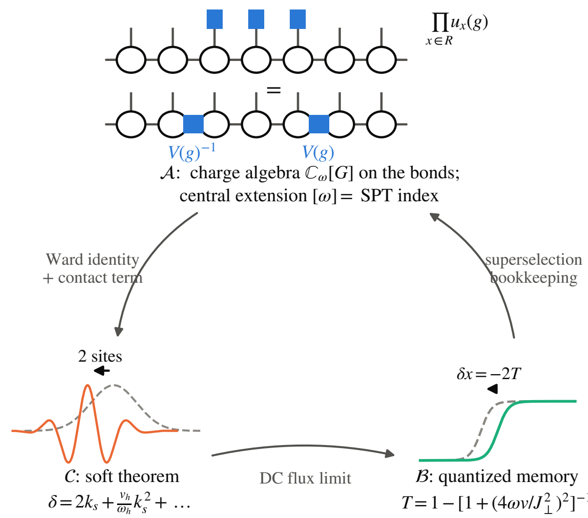

FIG. 1. The infrared triangle at finite lattice spacing. Corner $\mathcal{A}$ (asymptotic symmetry, proved): a symmetry applied to a region of an MPS collapses to two bond operators $V(g)^{-1}, V(g)$ at the region's ends; on windows padded about a bond these generate the charge algebra $\mathbb{C}_\omega[G]$, whose central extension $[\omega]$ is the SPT index. Corner $\mathcal{C}$ (soft theorem, proved for two-body scattering in the bilinear isotropic ferromagnet, hard momentum inside $(0, \pi)$, band edges and equal velocities excluded): a soft magnon exits with phase slope 2 at spin $\frac{1}{2}$, i.e. displaced by two sites ($1/s$ at site spin s). Corner $\mathcal{B}$ (memory, proved conditional on the scattering hypothesis H-AD-G: complete kink–magnon wave operators for a fixed packet with no bound component, one reflected and one transmitted channel of definite charges ∓ 1, local decay, limits taken in the order volume, time, window): a transmitted magnon displaces the wall by exactly $-1/s$ sites, a reflected one by zero. No edge is a theorem: the $\mathcal{A} \Rightarrow \mathcal{C}$ edge remains Conjecture S at every order—Corner $\mathcal{C}$ itself is proved two-body; $\mathcal{C} \Rightarrow \mathcal{B}$ passes soft data to memory only through the transmission probability; $\mathcal{B} \Rightarrow \mathcal{A}$ is charge bookkeeping within a fixed kink sector.

Setting.—An infinite chain of finite-dimensional sites; a compact symmetry group G acting on-site by $u(g)$; a finite-range G-invariant Hamiltonian H; and a family of translation-invariant injective matrix-product ground states with tensors A_α and bond dimension χ, one for each vacuum α, permuted by the symmetry as $\alpha \mapsto g \cdot \alpha$. One structural fact is used throughout, the fundamental theorem of MPS symmetries [15, 16]: acting with $u(g)$ on the physical index of an invariant A_α equals a phase times $V(g)^{-1} A_\alpha V(g)$, with V a projective bond representation whose class $[\omega] \in H^2(G, U(1))$ is the symmetry-protected topological (SPT) index [15, 17]. All theorems below are stated in this $(G, \text{injective MPS}, \text{finite range})$ register; Appendixes A–D linearize the proofs, with the repository's shards and machine-checked certificates as ground truth (Appendix E).

Asymptotic symmetry.—Apply the symmetry to a finite region R only: $U_R(g)=\prod_{x\in R}u_x(g)$. The intertwining relation, used once per site, cancels the V's pairwise through the interior, and the whole operation collapses to two bond insertions, $V(g)^{-1}$ entering R and $V(g)$ leaving it (Fig. 1). Everything a truncated symmetry does, it does at its two ends. Sending an end to infinity makes this an asymptotic-symmetry statement; three facts hold (Appendix A). (i) On states, a half-infinite symmetry string acts *exactly* as a single bond insertion; as an operator sequence it converges strongly if and only if $V(g)$ is a phase—otherwise there is no strongly convergent implementer at all, the lattice counterpart of a large gauge transformation. (ii) On finite windows padded around the distinguished bond, the insertions compose as the twisted group algebra $\mathbb{C}_\omega[G]$: the SPT class is precisely the central extension of the asymptotic charge algebra—a group-cohomological multiplier, not a Lie-algebra central charge (Appendix D). Whether these window states extend to normal states of the physical half-infinite algebra is open—hypothesis H-split; no statement below uses it silently. (iii) If the symmetry is broken, the half-infinite string leaves the vacuum sector: every finite truncation remains in the vacuum folium (it carries a kink–antikink pair at its two ends), and only the limit—in expectation values, never in norm—lands in a *domain-wall* (kink) sector. Kinks are what broken asymptotic symmetries create. When G acts transitively on the vacua, kink sectors are labelled by ordered vacuum pairs, with double-coset invariant in $H_\alpha\backslash G/H_\alpha$ (H_α the unbroken subgroup).

The same telescoping yields a lattice Noether identity: on the vacuum, for a normal-ordered *unbroken* generator, the physical charge density equals the divergence of a purely virtual bond quantity; separately, for *any* generator and any finite-range invariant H the deformed charge Q_k obeys the exact continuity equation $[H,Q_k]=(\mathrm{e}^{\mathrm{i}k}-1)J_k$. The kinematic factor is a property of the profile, not a soft factor; the soft theorem must be earned from the current.

The soft theorem.—Fix the concrete model: the spin-$\frac{1}{2}$ Heisenberg ferromagnet $H=-J\sum_x(\mathbf{S}_x\cdot\mathbf{S}_{x+1}-\frac{1}{4})$. Its polarized ground state is an exact $\chi=1$ MPS, magnons are one-particle eigenstates with $\omega(k)=J(1-\cos k)$ and velocity $v(k)=J\sin k$, and $SU(2)$ breaks to $U(1)$, making the magnon a type-B Goldstone mode [9]. Write $S_{12}=\mathrm{e}^{\mathrm{i}\delta}$ for the exact two-magnon amplitude, δ the continuous phase branch vanishing with the soft momentum k_s, at fixed hard momentum k_h.

Theorem 1 (soft magnon theorem, two-body)**.** *For k_h in a compact subset of $(0,\pi)$ and signed $k_s\to 0$,*

$$\delta \;=\; 2\,\mathrm{sgn}(v_h-v_s)\,k_s \;+\; \frac{|v_h|}{\omega_h}\,k_s^2 \;+\; O(k_s^3) \qquad (1)$$

uniformly, with $v_h=v(k_h)$, $\omega_h=\omega(k_h)$.

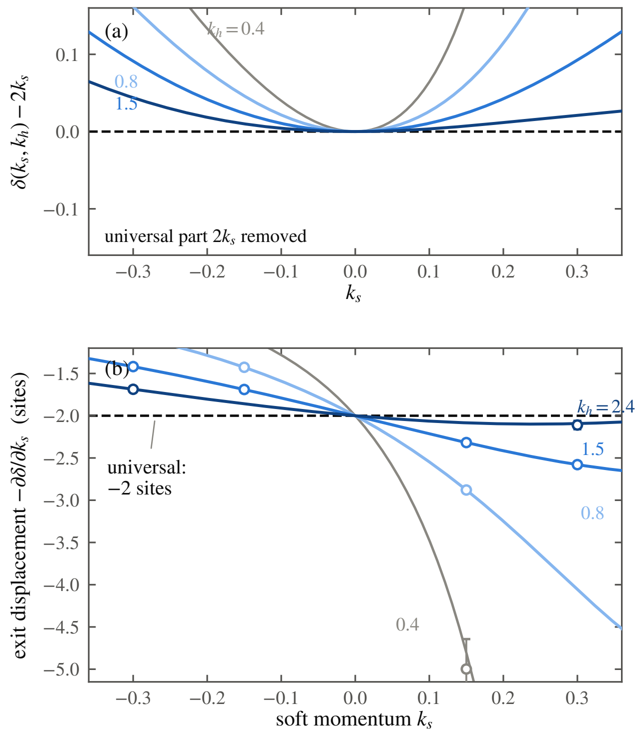

FIG. 2. The soft magnon theorem and its verification. (a) The exact two-magnon phase of the Heisenberg ferromagnet with its universal part removed: $\delta(k_s,k_h)-2k_s$ vanishes quadratically for each hard momentum k_h in a compact subset of $(0,\pi)$—the hard leg enters only at order k_s^2, through the magnon frequency $\omega_h=J(1-\cos k_h)$ and velocity $v_h=J\sin k_h$, with coefficient v_h/ω_h, Eq. (1). (b) The displacement reading: the soft packet's exit displacement $-\partial\delta/\partial k_s$ for the same k_h range (curves, exact) passes through -2 sites at $k_s=0$. Circles: real-time wavepacket measurements, which reproduce the universal coefficient 2 to 0.2%. The gray curve ($k_h=0.4$) shows the non-uniformity of the expansion as $k_h\to 0$; the universal -2 survives it. Every soft magnon leaves a two-site footprint, whatever magnon it scatters off—within two-body scattering; beyond is Conjecture S, unproved.

The zeroth order, $S_{12}\to 1$, is soft-magnon decoupling (a zero-momentum magnon is a symmetry rotation of the vacuum); the content is the next two orders. The linear coefficient is the pure number 2, up to the sign that records which particle is faster: independent of the hard momentum and of the coupling on the stated compact domain. Hard data enters first at second order, through the single even invariant $|v_h|/\omega_h$. And the 2 is not a scattering length—the ferromagnet's scattering length vanishes [12]—it is a *displacement*: the soft packet exits the collision shifted by two lattice sites (Fig. 2). The derivation (Appendix B) does not solve the model: one contact relation plus the implicit function theorem fixes the linear coefficient, all hard dependence cancelling. It has a Ward reading: the residue $\langle k_h|Q_0^\dagger J_0^-|k_h\rangle=2\mathrm{i}\,v_h$ is charge times velocity, mirroring the continuum soft-current analysis [3, 10]. Bethe integrability enters only as an oracle: the closed-form expansion reproduces Eq. (1) term by term, and a directly diagonalized two-magnon completeness theorem—not assumed—makes the expansion unconditional. Exact diagonalization confirms the form factors to 2×10^{-14} and both coefficients to 3×10^{-10}; wavepacket collisions reproduce the displacement, the coefficient 2 to 0.2%.

How far beyond two bodies does this reach? The strongest objection first: universality *fails* for unrestricted local sources—an explicit four-site operator O_η creates the same one-magnon state yet shifts the linear soft coefficient by $2\mathrm{i}\eta(1-\mathrm{e}^{-3\mathrm{i}k_h})$ (Appendix B). We conjecture—Conjecture S—that for n-particle scattering on an injective MPS vacuum, with sources in the five-condition Ward-covariant no-contact class $\mathcal{S}_W$ of Appendix B, $M_{n+1} = \mathsf{S}(k_s)\, M_n + o(k_s)$ in wavepacket norm, the multiplier S depending only on the legs' asymptotic charges and velocities, linear at leading order. Theorem 1 is its proved two-body instance, $\mathsf{S} = 2\mathrm{i}k_s$. All else is open—wave operators, form-factor regularity, the soft-limit interchange, two or more hard legs, membership of any microscopic source in $\mathcal{S}_W$; Appendix B states the complete list.

Memory.—Memory needs a wall, so change the model minimally: the easy-axis XXZ ferromagnet, anisotropy $\Delta > 1$, with two product vacua and a domain wall between them; send in a magnon wavepacket and ask where the wall sits afterwards (Fig. 3). Read the wall position from a window W around it, $\mathfrak{X}_W = \mathrm{const} + \frac{1}{2s}\sum_{x\in W} S^z_x$. Its change over the event is an exact operator identity, no hypotheses: the time-integrated $U(1)$ current through the two boundary bonds of W, i.e. the zero-frequency weight of the physical boundary current—as in the continuum, memory sits in the field at the boundary [2, 7]. Charge bookkeeping then gives the memory law, in the general register.

Theorem 2 (memory law)**.** *Let G be compact, $\{A_\alpha\}$ a covariant family of injective MPS vacua of any bond dimension, H finite range and G-invariant; fix a kink sector between two vacua carrying charge density $\pm s$ along a common unbroken $U(1)$, and let the incoming state be one fixed kink-magnon wavepacket in that sector whose charge deviations from the two vacua are summable—the ℓ^1 kink class (no plane wave). Assume the scattering hypothesis H-AD-G (Appendix C) for this packet: wave operators exist and are complete on its sector; the outgoing space carries exactly one reflected and one transmitted channel, of charges $q_{\mathrm{in}} = q_L = -1$ and $q_T = +1$ relative to their supporting vacua, and no further propagating channel; the packet has no bound-state component; local observables decay on the moving legs; and the limits are taken in the order infinite volume, then time, then observation window. Then the wall-displacement operator is*

$$\Delta X \;=\; -\frac{1}{s}\,N_T, \qquad \delta x \;=\; -\frac{1}{s}\,\langle N_T\rangle\,, \tag{2}$$

with N_T the transmitted-channel projection: each transmitted magnon displaces the wall by exactly $-1/s$ sites, each reflected one by zero, and $\mathrm{spec}(\Delta X) \subseteq \{-1/s, 0\}$ *with Bernoulli variance.*

The displacement is quantized by charge conservation alone: momentum, anisotropy, packet shape, and scattering phase enter only through $\langle N_T\rangle$. The proof is charge bookkeeping (Appendix C). One hypothesis is structural: the two vacua must carry *opposite* charge density along one unbroken direction; the spin flip supplies this in XXZ, and a bare abelian symmetry cannot. H-AD-G itself we prove only for the dominant domain-wall sector of the XXZ chain, conditional on an explicitly stated sector-reduction hypothesis; for the full chain it remains an assumption (Appendix C). Sparse-sector numerics at $N = 160$ confirm $\delta x = -2\,\langle N_T\rangle$ at $s = \frac{1}{2}$ to 0.064 sites across the scan (worst for the slowest packets, trapped weight $\approx 4\times10^{-3}$), and to 0.005 sites on the nine rows with trapped weight below 10^{-6}.

In the reduced sector the kink-magnon problem is a side-coupled (Fano) level, giving the closed form $t(k) = [1 + \mathrm{i}J^2/4\omega(k)v(k)]^{-1}$ with $\omega(k) = J(\Delta - \cos k)$; the measured reflection matches this over $\Delta \in [1.5, 12]$ to 0.9–5.8%, no fitted parameter (criterion 8%, fixed in advance). As $k \to 0$,

$$T(k) = 16(\Delta-1)^2k^2 + O(k^4), \tag{3}$$

so a soft magnon is totally reflected and leaves no memory: the memory effect has an Adler zero of its own. Whether the zero's coefficient is universal we have not established; that remains a conjecture.

Now the refutation. The continuum dictionary—memory = DC limit of the soft factor, summed over the event—is *false* in the XXZ chain, twice over: the displacement per transmission is fixed by charge conservation whatever the scattering phase does, and a purely transmitting wall with identically zero phase shift still displaces by exactly $-1/s$. The soft factor is a phase; the memory quantum is a charge. What survives—and is proved—is the pair above: memory is the DC weight of the boundary current, and soft data reaches it only through the transmission probability: charge transport, not phase accumulation.

The remaining edge is bookkeeping, and only that: under finite-range dynamics the vacuum-pair label of the kink sector is rigid at finite times, and $2s\,\delta x + (q_{\mathrm{out}} - q_{\mathrm{in}}) = 0$ within that fixed sector. Whether the memory *reconstructs* the asymptotic transformation which acted remains open.

One invariant, read out twice.—Two 2's appear above: the soft phase slope of Theorem 1 and the spin-$\frac{1}{2}$

memory quantum $1/s$ of Theorem 2. We conjecture—Conjecture B—that both equal $|q_{\rm hard}|/s$, the hard particle's $U(1)$ charge in units of the vacuum spin density. It was built to fail at $s \neq \frac{1}{2}$, where $1/s$ and the constant 2 separate; it survived (Fig. 3c, inset). Two independent implementations measured the phase slope $= 1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ and the memory quantum $= -1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}\}$, within pre-registered 8% bands (worst gated deviation 2.7%). An exact two-magnon computation now proves the slope half for the bilinear isotropic ferromagnet at every site spin: on the regular two-magnon domain, at each fixed hard momentum in $(0, \pi)$ and fixed channel, $\partial\delta/\partial k_s|_0 = \mathrm{sgn}(v_h - v_s)/s$ in closed form, hard dependence cancelling (Appendix B)—the unit-charge two-body slope only, with band edges and equal velocities excluded. The memory half remains conditional on H-AD-G; the $|q_{\rm hard}|$ factor is untested—every leg so far carries $|q| = 1$—and a charge-2 bound-state projectile is the next falsifier.

Where the SPT index lives.—Corner $\mathcal{A}$ made $[\omega]$ the central extension of the soft charge algebra; does it show up in soft amplitudes? The answer, proved in Appendix D, is a rigidity dichotomy. In the *bulk*, a closed symmetry insertion has two endpoints whose projective multipliers cancel exactly, and every normalized bulk coefficient is continuous along common-gap symmetric deformation paths with continuously varying external data (a coefficient reached through a soft limit or p derivatives needs in addition C^p external data and the uniformity hypothesis H-soft-p, Appendix D): nothing bulk is a topological invariant without a separate local-constancy proof. Along a symmetric path through the AKLT phase, a natural bulk soft-charge coefficient moves from 0.125 to 0.240 with the SPT class fixed. At an *unpaired endpoint* the situation inverts: the compensated soft residue is an operator on the χ-dimensional edge register whose centered charge has spectrum in the $[\omega]$-shifted lattice $q^\circ_\omega + \mathbb{Z}$ and dimension at least the minimal ω-projective dimension. For the AKLT half-chain the residue is exactly $-\frac{1}{2}[1-(2b^2-1)^L]\, Z \to -Z/2$ along the entire path: eigenvalues $\pm\frac{1}{2}$, rigid, while the bulk coefficient drifts. The topological data of soft physics concentrates at the boundary. Physical edge statements carry the hypothesis H-split above; granted in addition an edge-resolved scattering hypothesis H-AD-edge, charge conservation, and definite channel charges, charge bookkeeping quantizes edge-memory outcomes exactly as in Theorem 2: what is protected is the memory's *capacity*—a projective multiplet no symmetric perturbation can split—not any particular amplitude. That an AKLT boundary magnon actually writes to it, we state as a conjecture with a definite Hamiltonian and a pre-registered test (Appendix D).

Outlook.—The chain's topological data thus concentrates in a boundary soft algebra, $\mathbb{C}_\omega[G]$ at the ends, in structural parallel to the celestial program's symmetry towers at the boundary of spacetime [1]; its symplectic origin is deferred to forthcoming work. The wall is a broken asymptotic symmetry made visible, and—if Conjecture B holds—the magnon's soft phase and the wall's memory read out one conserved charge crossing an observer. The infrared structure of gauge theory and gravity, long tied to null infinity, fits on a chain of spins.

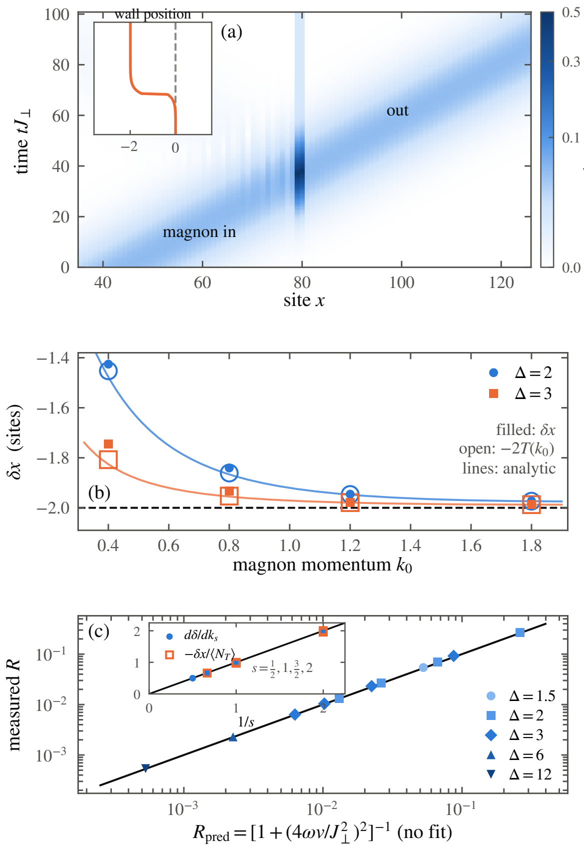

FIG. 3. The quantized memory effect in the XXZ chain ($\Delta > 1$). (a) Space–time record of one event ($\Delta = 3$, $k_0 = 1.2$, $N = 160$): flipped-spin density as a magnon wavepacket crosses the wall. Inset: the wall position steps back by two sites and stays—the permanent record. Across the plotted scan $\delta x = -2\,\langle N_T\rangle$ ($\langle N_T\rangle$: transmitted weight) holds to 0.064 sites (worst: slowest packets, trapped weight $\approx 4 \times 10^{-3}$); on the nine rows with trapped weight $< 10^{-6}$, to 0.002 sites. (b) Displacement versus momentum: filled δx, open $-2T$, curves the parameter-free Fano prediction. (c) Measured reflection against $R = [1 + (4\omega v/J^2)^2]^{-1}$, $\omega = J(\Delta - \cos k)$, $v = J \sin k$: collapse over $\Delta \in [1.5, 12]$, $k_0 \in [0.4, 1.8]$ to 0.9–5.8%; as $k_0 \to 0$ transmission vanishes quadratically, Eq. (3): a soft magnon leaves no memory. Inset: the spin-s falsifier of Conjecture B—soft phase slope (filled, rings up to 480 sites) and memory quantum $-\delta x/\langle N_T\rangle$ (open: every physical wavepacket run with trapped weight $< 10^{-2}$, the record's quality gate; controls excluded) against the line $1/s$, for $s = \frac{1}{2}, 1, \frac{3}{2}, 2$ (memory: $s \leq \frac{3}{2}$). Both are consistent with $1/s$, excluding the constant 2; that both measure one charge datum $|q_{\rm hard}|/s$ is Conjecture B.

We thank D. Giulini for discussions.

Corner A: the asymptotic charge algebra

Ground truth: repository shards `theory/corner-a.md`, `corner-a-kinks.md`, `corner-a-goldstone.md` (claim rows WI, A1, A2, G0, all PROVED through the adversarial verification loop recorded there); numerical certificates in Appendix E. Throughout, α labels vacua, $H_\alpha = \{g : g \cdot \alpha = \alpha\}$ the unbroken subgroup, and phases are normal-ordered so that the extensive phase of the truncated symmetry is removed.

Truncated symmetry (WI). For a finite region $R = [a, b]$ and any $g \in G$, exactly and with no limit, $U_R(g)\,|\psi(A_\alpha)\rangle = \mathrm{e}^{\mathrm{i}|R|\theta_\alpha(g)}\,|\psi(T_R^{(g)})\rangle$, where $T_R^{(g)}$ carries A_α outside R, $A_{g\cdot\alpha}$ inside, and two bond insertions: $V_\alpha(g)^{-1}$ on $\partial_- R$ and $V_\alpha(g)$ on $\partial_+ R$. The window must contain one site beyond each cut (or carry explicit edge-bond insertions); with interior bonds only, the identity is false, and the orientation of the two insertions matters—both failure modes are exhibited numerically in the certificate.

Unbroken case (A1). (a) Half-infinite strings act on states exactly as single bond insertions, and the convergence is eventually constant on every local algebra, not asymptotic. (b) They are implemented by a strongly convergent operator sequence if and only if $V_\alpha(g)$ is scalar. (c) Two insertions give the same state iff they differ by a scalar, so the endpoint states form a $PGL(\chi)$-torsor. (d) On windows padded about the bond the map $M \mapsto$ insertion is injective, and the insertions realize a linear representation of the twisted group algebra $\mathbb{C}_{\omega_\alpha}[G]$ in which the multiplier acts nontrivially; padding is necessary—an explicit $\mathbb{Z}_2$-symmetric counterexample shows the unpadded action is ill defined. On states the multiplier dies and what remains is a homomorphism $G \to PGL(\chi)$; what $[\omega_\alpha]$ obstructs is lifting it to an honest unitary representation. (e) *Open (H-split):* that these endpoint states are normal states of the physical half-infinite algebra—equivalently that $\mathbb{C}_\omega[G]$ acts on a physical edge Hilbert space—is not proved; the expected route is the split property. Every physical-edge statement in this Letter carries this hypothesis explicitly.

Broken case (A2). For $g \notin H_\alpha$, every finite truncation stays in the vacuum sector (a kink-antikink pair at the two ends); the half-infinite limit exists in the weak-$*$ sense only, with an exponential rate, and lands in the kink sector $\mathcal{K}_{\alpha, g\cdot\alpha}$, disjoint from the vacuum folium. The jump is invisible at every finite truncation and created entirely by the surviving end. Under the transitivity hypothesis (T), vacuum pairs form $(G/H_\alpha) \times (G/H_\alpha)$ and the complete diagonal invariant is the double coset in $H_\alpha \backslash G/H_\alpha$; for $G = SU(2)$ broken to $U(1)$ this is the relative angle of the two vacuum axes. Without (T) the classification holds per orbit. Two load-bearing gaps are recorded in the shards and respected here: H-split above, and the absence of a g-uniform statement when the vacuum manifold is continuous.

Currents (G0). For any generator ξ and finite-range invariant H: $[H, q_x(\xi)] = j_{x-1|x}(\xi) - j_{x|x+1}(\xi)$ with the local cut current, and in wavepacket form $[H, Q_k] = (\mathrm{e}^{\mathrm{i}k} - 1) J_k$. Separately, for a normal-ordered *unbroken* generator $\xi \in \mathfrak{h}_\alpha$ only: on the vacuum, the charge density acts as the lattice divergence of a virtual bond quantity—the potential is the fundamental object, as a theorem. The finite-window Fourier identity carries two $\Theta(1)$ boundary terms; the clean $(1 - \mathrm{e}^{\mathrm{i}k})$ form holds only in the stated δ-normalized sense. The kinematic factor supplies no soft theorem (a withdrawn overclaim is recorded as a refuted row in the claims ledger).

Corner C: the two-body soft theorem and its limits

Ground truth: `theory/soft-current-recon.md` (row S2-2body, PROVED), `theory/oracle-bethe.md` (oracle facts O1–O10, verification role only), `theory/ml2-completeness.md` (row ML2, PROVED), `theory/spin-s-twomagnon.md` (row S2-2body-S, PROVED through its adversarial verification round), `theory/ml4-ward-reduction.md` and `theory/ml5-universality.md` (rows ML4-A, ML4-Ward, ML5-A/B PROVED as conditional implications; ML5 unrestricted REFUTED).

Derivation of Theorem 1. The exact current of the model is $j^-_{x,x+1} = \frac{J}{2}(S^-_{x+1} - S^-_x) P_{x,x+1}$. A two-magnon eigenwave is free away from contact; writing the incoming coefficient as 1 and the outgoing as $s(k_s, k_h)$, the single contact bond imposes

$$(2z_h - z_s z_h - 1)\, s + (2z_s - z_s z_h - 1) = 0, \qquad z = \mathrm{e}^{\mathrm{i}k}, \quad (4)$$

whose unique analytic solution near $k_s = 0$ has k_s-coefficient $2\mathrm{i}$, all hard dependence cancelling; the continuous logarithm gives Eq. (1), with remainder uniform on compact hard sets. The Ward reading: the state $Q_0\,|k_h\rangle$ is the Bethe descendant with one rapidity at infinity, and $\langle k_h | Q_0^\dagger J_0^- | k_h\rangle = 2\mathrm{i}v_h$; dividing a hard external-leg reduction by the energy shift $v_h k_s$ cancels the velocity and leaves the 2. Symmetry alone does not prove the theorem: the current has an orthogonal contact component, and what makes it subleading in the two-body sector is energy-shell matching plus C^1 regularity of the on-shell trace (an abstract cancellation lemma proved in the shard), not the Ward identity. Completeness of the two-magnon resolution—scattering states, one bound band, descendants, and one singular vector on even rings—is proved by direct Jacobi diagonalization; no Bethe completeness is assumed, and integrability appears nowhere in the hypotheses.

The universality boundary. With the four-site local operator $D = S_0^- S_1^- - S_1^- S_2^- + S_2^- S_3^- - S_0^- S_3^-$, the source $O_\eta = S_0^- + \eta D$ creates the same one-magnon amplitude as S_0^- but shifts the linear soft coefficient of the induced two-body amplitude by $2i\eta(1-e^{-3ik_h})$. Unrestricted process independence is therefore refuted. The repaired statement (proved as a conditional implication) holds on the source class $\mathcal{S}_W$ defined by five conditions—an exhaustive normed LSZ decomposition, Ward covariance of the descendant current residue, a process-independent C^1 kinematic LSZ normalization, reduced-channel regularity with relative norm control, and no direct soft contact—and Conjecture S, in the form $M_{n+1} = \mathsf{S}(k_s)M_n + o(k_s)$ with S a function of the asymptotic leg charges and velocities only, asserts universality on exactly this class. Open, and stated as such: infinite-volume wave operators (ML1), uniform form-factor regularity including the $k = \Theta(1/N)$ regime (ML3), the packet-smeared infinite-volume one-hard-leg estimate and the $n \geq 2$-hard channel (ML4; the fixed-volume formulas are an off-shell interpolation whose naive volume-uniform reading is refuted), the exhaustive LSZ decomposition itself, nonemptiness of and microscopic membership in $\mathcal{S}_W$, and the limit-order control (ML6).

Spin-s slope (theorem). For the ferromagnet $H_S = -J\sum_x(\mathbf{S}_x \cdot \mathbf{S}_{x+1} - S^2)$ at every site spin S, the separated, adjacent, and (for $S \geq 1$) double-occupancy equations directly determine the closed-form regular two-magnon amplitude, with no integrability or completeness assumption; for each fixed $0 < |k_h| < \pi$ the physical phase has $\partial_{k_s}\delta_{\mathrm{phys}}|_0 = \mathrm{sgn}(v_h - v_s)/S$, locally uniformly on compact hard subsets with fixed channel (row S2-2body-S, PROVED). The scope is exact: this proves the unit-charge two-body slope only—not endpoint or equal-velocity limits, not spin-S Bethe completeness, not Conjecture S, not the memory half, not the $|q_{\mathrm{hard}}| > 1$ factor, and not Conjecture B.

Corner B: flux identity, memory law, Fano reduction

Ground truth: `theory/memory-quantization.md` (rows M-flux PROVED; M-quant PROVED conditional on the hypothesis D18, the scattering content of H-AD-G; Mq-AD3 PROVED conditional on the sector-reduction hypothesis Mq-E), `theory/memory-quantization-general.md` (row M-quant-G, PROVED as a conditional implication), `theory/corner-b-draft.md` (rows K1–K3, B3 PROVED; K4 CONJECTURE; row M REFUTED).

Flux identity (exact). For any state, any window $W = [a,b]$, and any finite times, $\frac{d}{dt}\mathfrak{X}_W$ equals $\frac{1}{2s}$ times the difference of the two boundary-bond currents; integrating, $\delta x = \frac{1}{2s}[\tilde{\jmath}_{a-1|a}(0) - \tilde{\jmath}_{b|b+1}(0)]$, the zero-frequency weight of the physical boundary current. No kink, spectral, or asymptotic hypothesis enters. (An earlier reading of this identity through virtual bond data was deleted during verification; the current here is physical.)

Hypothesis H-AD-G. For one fixed normalizable kink-magnon wavepacket, wave operators $W_\pm$ exist and are complete on its sector; the outgoing space splits into exactly one reflected and one transmitted channel with definite charges $q_{\mathrm{in}} = q_L = -1$, $q_T = +1$, and no further propagating channel; the packet has no bound-state component; local observables decay on the moving legs; and limits are taken in the stated order (infinite volume, then time, then window). For Theorem 2 in the general register these charges, the density jump $2s$, and summable charge deviations from the two vacua (an ℓ^1 kink class) are the only inputs: conservation of the regularized charge $2s(\mathfrak{X}_W - c)$ plus leg bookkeeping gives $2s\,\delta x + (q_{\mathrm{out}} - q_{\mathrm{in}}) = 0$ per channel, hence Eq. (2); the quantum for a charge-ν transfer is $-\nu/2s$. The structural remark in the text is the shard's: covariance forces $\omega_{g\cdot\alpha}(\xi) = \omega_\alpha(\mathrm{Ad}_{g^{-1}}\xi)$, so if the broken element commutes with the unbroken direction ($\mathrm{Ad}_{g^{-1}}\xi = \xi$, as for any abelian G) the two vacua carry *equal* densities and no such kink pair exists; the required inversion is supplied by a Weyl-type element, as the spin flip in XXZ. Nothing in compactness, injectivity, or finite range implies H-AD-G itself; it is proved here only where stated next.

Sector reduction and the Fano level. In the XXZ chain at $s = \frac{1}{2}$ the incoming component with at most three domain walls, under the enumeration hypothesis Mq-E (verified explicitly at $N = 14$; all-volume form open), is unitarily a two-sided tight-binding chain with one side-coupled level—the flat kink state, whose bondwise annihilation by the exact kink Hamiltonian is proved. Kato–Rosenblum and a Feshbach analysis then give H-AD-G's scattering clauses for this reduced dynamics: trace-class perturbation, complete wave operators, no singular-continuous spectrum. Eliminating the side level yields $t(k) = [1 + iJ^2/4\omega v]^{-1}$ and Eq. (3). Leakage of the full chain out of the reduced sector is measured at $O(\Delta^{-2})$ (probabilities 0.10/0.03/0.008 at $\Delta = 2/4/8$); it affects $T(k)$, not the conservation laws. Full-chain H-AD-G and the universality of the soft-zero coefficient remain open and are tracked as such in the claims ledger.

Refutation of the naive dictionary. The brief's Conjecture M ("δx is the DC limit of the soft factor") fails on two counts: the channel quantum is independent of the scattering phase once the channels exist, and a phaseless purely transmitting wall still displaces maximally. The refuted row is retained in the ledger with its counterexamples; the surviving pair is the flux identity plus Eq. (2).

Label rigidity. Under stationary vacua and finite-range (Lieb–Robinson) dynamics the factorized vacuum-pair boundary label of the kink sector is preserved at all finite times, so δx is motion within a fixed sector, not a change of sector. The thermodynamic uniqueness/flatness of the kink family (a $\mathbb{Z}$-torsor reading) remains a conjecture with finite-volume evidence.

The SPT rigidity dichotomy

Ground truth: `theory/spt-rebuild.md` (rows SPT-B-mult, SPT-B′, SPT-E-AKLT, SPT-E′, SPT-T′, SPT-D′, SPT-M′ all PROVED with the hypotheses below explicit; SPT-M′-dyn CONJECTURE; the earlier pointwise bulk-blindness claim and the all-orders edge no-go both REFUTED and retained as negative rows).

Bulk (deformable). A closed on-site insertion leaves $V(g) \otimes \bar{V}(g)$: the multipliers cancel exactly, for every lift, so no closed-bulk amplitude carries a projective anomaly. This does not make bulk data blind to the phase—adjoint representation content distinguishes Pauli-projective from scalar conjugation—but it is deformable: along any common-gap symmetric path with continuous external data (tensors, channel embeddings, gauge fixes, normalizations), every normalized bulk coefficient is continuous; a coefficient reached through a soft limit, or through p derivatives, requires in addition C^p external data and the uniformity hypothesis H-soft-p of the shard. On the anisotropic AKLT family ($A^x_b \propto X$, $A^y_b \propto Y$, $A^z_b \propto Z$) the soft-charge coefficient is $C_{\text{bulk}}(b) = b^2/4(1-b^2)$: $\frac{1}{8}$ at the AKLT point, 0.2402 at $b = 0.7$, same phase, same $[\omega]$.

Edge (rigid). On the fixed χ-dimensional Schmidt register of a half-chain, the compensated endpoint residue of a symmetry string is $V(g)$ itself, and the Hermitian charge residue is the centered generator $Q_{\text{edge}} = -\mathrm{i}X^\circ$, lift-gauge invariant, with spectrum in the shifted lattice $q^\circ_\omega + \mathbb{Z}$ and every irreducible summand of dimension $\geq d_\omega$ (the minimal ω-projective dimension; > 1 whenever $[\omega] \neq 0$). For the AKLT family the finite-window residue is exactly $-\frac{1}{2}[1 - (2b^2 - 1)^L]Z$, with limit $-Z/2$ independent of b; the trivial comparison tensor gives 0. The half/integer offset is pinned by the global group ($O(2)$ here, or the $SU(2)$ cover); a bare $U(1)$ does not protect it.

Edge memory. Given H-split, the edge-resolved scattering hypothesis H-AD-edge, charge conservation, and definite channel charges, $\Delta Q_{\text{edge}} = -(Q_{\text{bulk,out}} - Q_{\text{bulk,in}})$, with channel outcomes in $\mathbb{Z}$ (for the AKLT doublet, in $\{-1, 0, 1\}$)—the same bookkeeping as Theorem 2 with edge charge in place of wall position. Protection is capacity protection: the projective multiplet cannot be split symmetrically. That the specific open AKLT parent Hamiltonian with boundary coupling $P^{(S=2)}_{0,1}$ has a nonzero edge-changing reflection amplitude on an open momentum interval is conjecture SPT-M′-dyn; the deciding half-chain scattering computation, with pre-registered acceptance gates, is specified in the shard. A null result there refutes that conjecture only, not the registered endpoint theorems.

Numerical certificates

The theorems above carry independent machine-checked certificates under `theory/checks/` (ten checkers, all passing; five of them—the Ward-reduction/universality, flux/memory-scan, general-memory, spin-S slope, and SPT checkers—carry documented failing “red” mutation modes), and the quantitative claims of the text trace to committed data files: exact-diagonalization form-factor residuals $\leq 1.6 \times 10^{-14}$ and coefficient fits to 2.2×10^{-10} (`soft_current_recon_check.py`); two-magnon completeness on rings up to 26 sites (`ml2_completeness_check.py`); the flux identity to 3.4×10^{-16} (`mquant_check.py`); the wavepacket memory scan (`memory-scan-1.json`: $N = 160$, sparse kink-sector propagation; $|\delta x + 2\langle N_T\rangle| \leq 0.064$ sites across the full momentum scan, dominated by the slowest packets with trapped weight $\approx 4 \times 10^{-3}$, and ≤ 0.005 sites for both estimators on the nine rows with trapped weight $< 10^{-6}$); the Fano cross-check (0.9–5.8% over $\Delta \in \{1.5, 2, 3, 6, 12\}$, pre-registered 8% criterion); the spin-s falsifier and its independent cross-check (`spin1-bc-falsifier.json`, `spin1-bc-crosscheck.json`: ansatz-free ring spectra to $N = 480$ plus real-time collisions, slope deviations from $1/s$ below 0.5% extrapolated; wavepacket memory ratios within 2.7% under the trapped-weight gate $< 10^{-2}$, within 1% at the cleanest kinematics); and the SPT deciding computation (`spt_rebuild_check.py`: bulk coefficient and edge residue to stated tolerances, sign and phase-gauge mutants failing). Figures 1–3 are generated from these committed files by `paper/figures/make_figures.py`; no number is hand-entered.

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